SmartCess

Complete Product, Design, and Technical Implementation Specification

Accessibility-first life-admin assistant for blind and low-vision users

Item	Details
Product	SmartCess Android application
Platform	Android
Primary users	Blind and low-vision adults
Core tester / co-designer	Hillarry
Product owner	Maurice
Primary AI provider	Meta Model API
Core AI models	Muse Voice Transcribe, Muse Spark
Supporting AI models	SAM 3.1, Muse Image
Development assistant	Muse Code
Modules	Daily Briefing, Bill Watcher, Email Briefing, Document Reader, Appointment Assistant, Routine Guardian
Document type	Product requirements, UI/UX design system, and technical implementation specification

1. Executive summary

SmartCess is an Android, voice-first, accessibility-first life-admin assistant for blind and low-vision users. It helps users independently understand and manage bills, email, letters and documents, appointments, medication, deliveries, and recurring tasks.

The product turns inaccessible information into simple spoken explanations, clear deadlines, and safe, user-approved actions. Its core promise is: **SmartCess turns inaccessible life admin into independent action.**

This document is the complete build specification for SmartCess. It defines product requirements, user journeys, screen inventory, navigation, UI and visual design, data model, Meta AI integration, guided scanning, prompts, structured output schemas, tool-calling contract, validation rules, state machines, accessibility requirements, error handling, offline fallback, security, performance, localization, analytics, testing, release process, and post-release roadmap.

The application must be usable with TalkBack from the first release. Voice is an additional interaction path, not a replacement for accessible touch navigation. Every major action must be available through both voice and touch.

Core product promise

SmartCess turns inaccessible life admin into independent action.

What SmartCess does

- Listens to the user through Muse Voice Transcribe.
- Understands intent and decides what to do through Muse Spark.
- Guides the user to position the phone correctly for scanning.
- Reads and analyzes bills, documents, email, and appointment information.
- Extracts providers, amounts, due dates, deadlines, and required actions.
- Explains information in simple spoken language.
- Compares current and previous bills.
- Creates reminders only after explicit user confirmation.
- Sends accessible notifications before deadlines.
- Answers follow-up questions using locally saved information.
- Provides an offline OCR and text-to-speech fallback when Meta AI is unavailable.

2. Problem statement

Many everyday administrative tasks assume that the user can see and interact with visual information. Blind and low-vision users often face barriers when dealing with printed utility bills, bank and telecommunication bills, government letters, medical appointment letters, email deadlines, PDF documents, medication schedules, delivery notifications, and recurring household tasks.

Existing tools may help with reading text or describing a scene, but they often do not help the user complete the next step: understanding what matters, deciding what to do, and creating a reminder or follow-up action. A further barrier is simply positioning the camera: a blind user cannot see whether the bill is centered, too far, too close, or tilted.

User pain points

- Bills are difficult to read, understand, compare, and track.
- Important email deadlines can be missed.
- Letters and PDF documents are hard to interpret independently.
- Appointment details, locations, and preparation instructions are hard to manage.
- Medication, deliveries, and recurring tasks are easy to forget.
- Users may need to ask sighted family members or support workers for help, reducing independence and privacy.
- Cloud-only AI tools may not work reliably in areas with poor connectivity.
- Users cannot see whether the document is framed correctly in the camera.

Why SmartCess is different

SmartCess is not merely a camera description app. It is a life-admin assistant. It helps the user move from information to action: understand a bill, identify a deadline, compare costs, prepare for an appointment, track medication, and create a reminder. It also guides the user physically through the scanning process with real-time spoken and haptic feedback.

Capability	General AI camera app	SmartCess
Read text	Yes	Yes
Describe surroundings	Yes	Yes
Extract bill details	Limited	Yes
Compare bills	Rare	Yes
Create reminders	Rare	Yes
Daily briefing	Rare	Yes
Appointment preparation	Rare	Yes
Routine tracking	Rare	Yes
Guided camera framing	Rare	Core feature
TalkBack-first design	Varies	Core requirement
Approval-gated actions	Rare	Core requirement

3. Users and personas

Primary persona: Hillarry

Hillarry is a blind Android user and SmartCess's primary tester and co-designer. He wants to independently manage bills, appointments, medication, and daily information. He needs clear spoken feedback, simple predictable workflows, and strong privacy protection, especially when dealing with financial documents, medical information, or personal items.

Secondary persona: rural or low-connectivity user

This user lives in an area with unreliable mobile data and uses an affordable Android phone. They need basic assistance even when internet access is limited and need an app that does not depend entirely on cloud services.

Secondary persona: busy blind professional

This user uses email, calendar, documents, and online services daily. They need fast summaries and deadline extraction, want to reduce reliance on sighted assistance, and value accuracy, privacy, and control.

Design implications

- Use simple, predictable navigation.
- Use large touch targets and high-contrast visuals.
- Make TalkBack the primary quality gate.
- Provide voice and touch parity for every major action.
- Explain AI uncertainty honestly.
- Require confirmation for actions.
- Store personal data locally by default.
- Provide an offline fallback.
- Provide real-time camera framing guidance for scanning.

4. Goals and success metrics

Product goal

Enable blind and low-vision users to independently manage everyday administrative tasks without needing sighted assistance.

Release goal

Deliver a reliable Android application that demonstrates all five SmartCess modules, with Daily Briefing and Bill Watcher as the deepest and most dependable workflows.

Success statement

SmartCess is successful when a blind user can independently ask what needs attention, scan a bill with guided camera framing, understand its details, compare it with a previous bill, create a reminder, receive a notification, scan or import a document, manage an appointment, and track a routine using TalkBack.

Release metrics

Metric	Target
Bill scan-to-result completion	Successful within 30 seconds
Guided framing success	User captures a well-framed bill within 3 guidance prompts
Bill extraction success	Successful on clear, well-lit bills
Reminder creation success	100% after user confirmation
TalkBack task completion	Core workflow completable independently
Core workflow reliability	No failure during primary demonstration flow
Meta AI usage	Muse Voice Transcribe and Muse Spark central to workflow
Module coverage	All five modules accessible and functional
Crash-free core workflow	No crashes during primary user journey

Post-release product metrics

Metric	Why it matters
Weekly active users	Measures ongoing usefulness
Bills scanned per user per week	Measures Bill Watcher adoption
Guided scan success rate	Measures whether framing guidance works
Reminders created per user	Measures action completion
Reminder completion rate	Measures whether reminders help users act
Daily Briefing usage	Measures return usage
Documents summarized per user	Measures Document Reader adoption
Appointments added per user	Measures Appointment Assistant adoption
Routine completion rate	Measures Routine Guardian effectiveness
Task completion without sighted help	Measures independence
TalkBack task success rate	Measures accessibility quality
User-reported trust score	Measures comfort with privacy and AI accuracy

5. Scope

In scope

- Daily Briefing.
- Bill Watcher: scan, extract, validate, review, compare, save, archive, remind.
- Email Briefing: summarize email, identify urgency, extract deadlines, create reminders.
- Document Reader: scan/import documents, summarize, extract deadlines, answer questions.
- Appointment Assistant: add appointments, extract details, create calendar event and travel reminder.
- Routine Guardian: medication, deliveries, recurring tasks, completion tracking.
- Guided scanning: real-time spoken and haptic camera framing feedback.
- Muse Voice Transcribe voice commands and Muse Spark intent routing.
- Muse Spark multimodal understanding, structured extraction, question answering, and action planning.
- TalkBack-first UI, local data storage, reminders, notifications, camera scanning, PDF/image import.
- Offline OCR and text-to-speech fallback.

Out of scope

- Automatic bill payment.
- Automatic email sending, replying, deleting, or forwarding.
- Automatic form submission or phone calls.
- Full navigation assistance.
- Medical diagnosis or medical, financial, or legal advice.
- Continuous background camera monitoring or facial recognition.
- Full control of third-party apps.
- Multi-user accounts, cloud sync, caregiver sharing, subscription payments.

6. System architecture

SmartCess is an Android application with Meta AI as its primary intelligence layer. The Android app captures user input, manages local data, executes confirmed actions, and presents accessible results. Meta models perform speech recognition, intent understanding, multimodal analysis, extraction, reasoning, and action planning. On-device ML Kit handles real-time framing detection and offline OCR.

High-level architecture

```
User voice / touch
        |
        v
Muse Voice Transcribe  (speech-to-text)
        |
        v
Muse Spark  (intent + reasoning + extraction + action planning)
        |
        +--> Bill Watcher
        +--> Email Briefing
        +--> Document Reader
        +--> Appointment Assistant
        +--> Routine Guardian
        |
        v
Validation + accessible confirmation
        |
        v
Room local database
        |
        v
WorkManager automation
        |
        v
Android notifications + TextToSpeech

Camera framing loop (on-device, real-time):
CameraX -> ML Kit text/document detection -> framing engine
        -> TTS + haptic feedback -> user adjusts phone
```

Technology stack

Layer	Technology	Purpose
Language	Kotlin	Primary Android development language
UI	Jetpack Compose + Material 3	Modern accessible Android UI
Architecture	MVVM + Repository	Separation of UI, domain, and data
Dependency injection	Hilt	Provides dependencies across app layers
Async	Kotlin Coroutines + Flow	Handles asynchronous AI and data operations
Local database	Room	Stores bills, documents, reminders, routines, and preferences
Background work	WorkManager	Schedules reminders and notifications
Camera	CameraX	Captures bill and document images
Framing detection	ML Kit Text Recognition / Document Scanner	Real-time text bounds detection for guidance
Haptic feedback	Android Vibrator / HapticFeedback	Non-visual framing cues
OCR fallback	ML Kit Text Recognition	Extracts text offline
Speech output	Android TextToSpeech	Speaks results, guidance, and confirmations
Voice input	Muse Voice Transcribe	Meta speech-to-text
Reasoning	Muse Spark	Meta multimodal reasoning and extraction
Visual highlighting	SAM 3.1	Optional region highlighting and framing refinement
Image generation	Muse Image	Optional visual explanations
Development	Muse Code	AI coding assistant
Networking	Retrofit/OkHttp or Ktor	Calls Meta Model API
JSON	Kotlinx Serialization	Parses structured AI output
Testing	JUnit, Turbine, MockK, Compose UI tests	Validates logic and UI

What Meta replaces

Meta replaces third-party speech-to-text, general-purpose language models, document-understanding services, and much of the custom natural-language logic. Muse Voice Transcribe handles speech input. Muse Spark handles intent classification, multimodal analysis, structured extraction, comparison, question answering, and action planning. SAM 3.1 provides optional region-level visual understanding.

What Meta does not replace

- CameraX remains required to capture images.
- ML Kit remains required for real-time on-device framing detection and offline OCR.
- Android UI frameworks remain required for the interface.
- TalkBack remains required for screen-reader accessibility.
- Room remains required for local storage.
- WorkManager remains required for reliable reminders.
- Android TextToSpeech remains required for spoken output.

7. Meta AI integration

AI responsibilities

Meta tool	Responsibility
Muse Voice Transcribe	Converts live speech or recorded audio into text
Muse Spark	Understands intent, analyzes images/documents, extracts structured data, summarizes, answers questions, compares bills, plans actions
SAM 3.1	Optional: region-level understanding of bill/document images; can support framing refinement and highlighting key regions such as amount or due date
Muse Image	Optionally creates simple visual explanations
Muse Code	Helps build, debug, and refine SmartCess

Core principle

Muse Voice Transcribe listens. Muse Spark thinks. ML Kit guides the camera. SmartCess acts safely.

Muse Voice Transcribe

Muse Voice Transcribe converts user speech into text. It supports realtime streaming and file transcription. SmartCess should use realtime streaming when the user taps Talk and speaks a command.

The realtime endpoint is a WebSocket endpoint for live audio, voice agents, dictation, and captions. SmartCess should capture mono 16-bit PCM audio, preferably at 24 kHz, and stream it to the endpoint. It should display interim transcripts where available and use the final transcript for intent classification.

Muse Spark

Muse Spark is SmartCess's brain. It is a multimodal reasoning model that can process text, images, documents, audio, and video, and supports tool calling and structured output.

Muse Spark must perform the following tasks:

- Classify user intent.
- Decide which SmartCess module handles the request.
- Analyze bill images and documents.
- Extract structured bill, document, appointment, and routine information.
- Compare current and previous bills.
- Summarize email and documents.
- Answer follow-up questions using local SmartCess data.
- Ask clarifying questions when information is missing.
- Propose safe next actions.
- Return structured JSON for validation.

Structured output

All critical Muse Spark outputs must use structured output. SmartCess must define a JSON Schema in the API request's response_format field so the model's output is constrained to match the schema.

Structured output is required for: intent classification, bill extraction, bill comparison, email briefing, document summary, appointment extraction, routine creation, and daily briefing.

Tool calling

Muse Spark should use tool calling to interact with SmartCess local data and actions. The app passes tool definitions to Muse Spark. When Muse Spark requests a tool, SmartCess executes it locally and returns the result to the model.

Tool	Purpose	Approval
get_today_data	Retrieve bills, appointments, routines, documents, and email items for briefing	No
get_bill_by_provider	Retrieve latest matching bill for comparison	No
get_bill_by_id	Retrieve a saved bill	No
save_bill	Save validated bill data	Yes
create_reminder	Create a reminder	Yes
cancel_reminder	Cancel an existing reminder	Yes
archive_bill	Mark a bill as paid or archived	Yes
save_document_summary	Save a document summary	Yes
create_appointment	Save an appointment	Yes
create_routine	Create medication, delivery, or recurring task	Yes
complete_routine	Mark a routine complete	Yes
ask_user_for_confirmation	Ask the user to approve an action	Yes

Approval model

SmartCess follows a prepare-confirm-act model. SmartCess may read, summarize, extract, and organize information automatically. It must never take an irreversible action without explicit user confirmation.

Action	Automatic?	Approval required?
Read a bill	Yes	No
Summarize email	Yes	No
Extract deadline	Yes	No
Save a bill	After review	Yes
Create reminder	No	Yes
Add calendar event	No	Yes
Send email	Not in MVP	Yes
Pay a bill	Not in MVP	Yes
Submit a form	Not in MVP	Yes

8. Guided scanning and camera framing

Guided scanning is a core accessibility feature. A blind user cannot see whether the bill or document is centered, too far, too close, tilted, or partially covered. SmartCess must provide real-time, simple spoken and haptic feedback to help the user position the phone correctly before capture.

Goal

Enable a blind user to capture a well-framed bill or document within a few guidance prompts, without sighted assistance.

How it works

- CameraX streams preview frames continuously while the Camera screen is open.
- ML Kit Text Recognition runs on-device on each frame (or every 2–3 frames to save battery).
- The framing engine analyzes the detected text block bounding boxes.
- Based on the analysis, SmartCess gives one simple instruction at a time via TTS and a short haptic pulse.
- When the document is well-framed, SmartCess says “Ready. Tap anywhere or say capture” and enables the Capture button.
- The user can also say “capture” by voice once framing is good.

Framing rules and feedback

Detected condition	Spoken guidance	Haptic cue
No text detected	“I do not see a document. Point the camera at the bill.”	Single short pulse
Document too far / text too small	“Move closer to the bill.”	Two short pulses
Document too close / text cut off	“Move back slightly.”	Two short pulses
Document off to the left	“Move the phone slightly to the left.”	One pulse
Document off to the right	“Move the phone slightly to the right.”	One pulse
Document too high in frame	“Move the phone down a little.”	One pulse
Document too low in frame	“Move the phone up a little.”	One pulse
Document tilted	“Hold the phone straight over the bill.”	Two pulses
Poor lighting / low contrast	“It is too dark. Move to better light if possible.”	Three short pulses
Well-framed and readable	“Ready. Tap capture or say capture.”	Long confirming pulse

Framing engine logic

- Compute the union bounding box of detected text blocks.
- Compare box center to frame center to determine left/right/up/down offset.
- Compare box width/height to frame to determine too far/too close.
- Estimate tilt from the angle of the dominant text baseline.
- Estimate lighting from average luminance of the text regions.
- Only speak a new instruction when the previous one is resolved or 3 seconds have passed (avoid spamming).
- Prioritize instructions: no text > too far/close > off-center > tilted > lighting.

Privacy and performance

- Framing detection runs entirely on-device using ML Kit; no frames leave the phone during guidance.
- Only the final captured image is sent to Muse Spark, and only after the user confirms capture.
- Process every 2–3 frames, not every frame, to protect battery.
- Stop the framing loop when the user leaves the Camera screen.

Optional SAM 3.1 enhancement

SAM 3.1 can optionally refine framing by identifying the document boundary more robustly (e.g., distinguishing the bill from the table background). This is a post-MVP enhancement; the MVP uses ML Kit only, so guided scanning works offline.

Acceptance criteria

- User receives spoken guidance when no document is detected.
- User receives “move closer” guidance when the bill is too far.
- User receives left/right/up/down guidance when the bill is off-center.
- User receives “ready” confirmation when framing is good.
- Guidance is also available as haptic pulses.
- Guidance works offline (ML Kit on-device).
- Guidance does not spam: one instruction at a time, with a pause between repeats.
- User can capture by voice (“capture”) or touch once framing is good.
- Framing loop stops when the user leaves the Camera screen.

9. Data model

SmartCess stores user data locally using Room. The database must support bills, reminders, documents, appointments, routines, email items, conversation history, and user preferences.

Entities and fields

- Bill: id, provider, amount, currency, dueDate, accountReference, billingPeriodStart, billingPeriodEnd, requiredAction, status, rawText, imageUri, confidence, previousBillId, amountChange, createdAt, updatedAt.
- Reminder: id, billId, title, message, remindAt, status, createdAt.
- Document: id, title, sourceType, fileUri, rawText, summary, keyPoints, deadline, amount, currency, requiredAction, confidence, createdAt.
- Appointment: id, title, providerName, location, startTime, endTime, doctorName, preparationNotes, questions, travelReminderAt, status, createdAt.
- Routine: id, title, type, schedule, nextDueAt, repeatRule, status, createdAt.
- EmailItem: messageId, sender, subject, receivedAt, summary, urgency, deadline, requiredAction, hasAttachment, isRead.
- ConversationTurn: id, sessionId, role, content, createdAt.
- UserPreferences: language, speechRate, defaultReminderTime, defaultReminderOffset, metaAiEnabled, emailConnectionStatus, calendarConnectionStatus, hapticFeedbackEnabled.

Data rules

- All personal data is stored locally by default.
- Every record must have a creation timestamp.
- Bills, documents, and reminders must support soft delete or permanent delete.
- The database must support schema versioning and migrations.

10. Module specifications

10.1 Daily Briefing

Daily Briefing gathers relevant information from bills, email, documents, appointments, and routines, then creates a short prioritized spoken briefing.

User flow

- User opens SmartCess or taps Talk and says: “What needs my attention today?”
- Muse Voice Transcribe converts speech to text; Muse Spark classifies intent as daily_briefing.
- SmartCess retrieves bills, appointments, email items, documents, and routines.
- Muse Spark creates a prioritized briefing; SmartCess speaks it.
- User selects an item by voice or touch; SmartCess opens the relevant detail screen.

Prioritization

- Overdue bills, bills due within seven days, appointments today or tomorrow, urgent email deadlines, documents with deadlines, medication and routines due today.

Example briefing

“Good morning. You have one appointment today at 2 p.m. Your electricity bill is due in three days. One email from your clinic asks you to confirm tomorrow’s appointment. You have one medication reminder at 8 p.m. Would you like to start with the bill or the appointment?”

Acceptance criteria

- User can request a briefing by voice or touch.
- SmartCess reads the briefing aloud and prioritizes urgent items.
- User can open a briefing item.
- The briefing works when some modules have no data and clearly states when nothing is urgent.

10.2 Bill Watcher

Bill Watcher helps users detect, understand, compare, remind, and archive bills.

User flow

- User says: “Scan my electricity bill.” Muse Voice Transcribe converts speech; Muse Spark identifies scan_bill.
- SmartCess opens CameraX and starts guided framing.
- SmartCess gives real-time guidance: “Move closer to the bill”, “Move slightly left”, “Hold steady”, then “Ready. Tap capture or say capture.”
- User captures the bill (by voice or touch); SmartCess sends the image to Muse Spark.
- Muse Spark returns structured bill data; SmartCess validates and shows the Bill Result screen.
- SmartCess reads the result aloud; user can save, rescan, discard, archive, or create a reminder.

Bill fields

- Provider, amount due, currency, due date, account reference, billing period start/end, required action, confidence score, missing fields, plain-language summary.

Bill comparison

If a previous bill from the same provider exists, SmartCess must compare amount, currency, due date, and billing period. It must explain whether the amount changed, by how much, the percentage change where meaningful, and whether the user should review the bill. It must not speculate about charges it cannot see.

Validation rules

Field	Rule
Provider	Must not be empty
Amount	Must be greater than zero
Currency	Must be valid or recognizable
Due date	Must be valid; warn if in the past
Required action	Must match allowed values
Confidence	Below 0.70 requires user review
Missing fields	Trigger clarification or rescan

Acceptance criteria

- User can scan a bill by voice or touch.
- Guided framing helps the user position the phone correctly.
- Muse Spark extracts bill details into structured information.
- SmartCess validates uncertain or missing fields.
- User can review extracted details with TalkBack and hear the summary.
- SmartCess compares with a previous bill from the same provider.
- User can save, archive, discard, or create a reminder.
- Reminder notification fires at the scheduled time.
- User can ask questions about saved bills.

10.3 Email Briefing

Email Briefing helps users understand urgent email without needing to read through a crowded inbox.

Implementation

For the initial release, implement Email Briefing using a local demonstration inbox or a simple email connection. If full email-provider integration is not reliable within the build period, use a local sample inbox containing realistic messages. The interaction model must still be complete: summarize, prioritize, extract deadlines, and create reminders with user approval.

User flow

- User says: “Brief me on my email.” Muse Spark identifies email_briefing.
- SmartCess retrieves recent unread messages; Muse Spark summarizes each, identifying urgency, deadlines, required actions, and attachments.
- SmartCess presents a spoken briefing; user can open a specific email summary or create a reminder from a deadline.

Email urgency

- Low: informational. Normal: no deadline. High: action requested or near-term deadline. Urgent: immediate deadline, critical notice, or time-sensitive requirement.

Safety rules

- Never send, reply to, delete, or forward email automatically.
- Never open links without user confirmation.
- Only create reminders after user approval.

Acceptance criteria

- User can request an email briefing by voice or touch.
- SmartCess summarizes unread email and identifies urgency, deadlines, required actions, and attachments.
- User can open a specific email summary and create a reminder from a deadline.
- SmartCess never performs irreversible email actions automatically.

10.4 Document Reader

Document Reader turns PDFs, letters, notices, and other documents into simple spoken summaries.

User flow

- User says: “Read this letter” and scans or imports a PDF.
- If scanning, SmartCess starts guided framing with the same real-time spoken/haptic guidance as Bill Watcher.
- SmartCess processes the document; Muse Spark produces a simple spoken summary.
- SmartCess identifies deadlines, amounts, names, and required actions.
- User asks follow-up questions and can create a reminder from a deadline.

Supported inputs and rules

- Camera-scanned printed documents, imported images, imported PDFs; single-page first, multi-page secondary.
- Summarize in plain language; extract deadlines, amounts, currency, required actions, sender.
- Do not provide legal, medical, or financial advice; clearly identify uncertain or missing information.

Acceptance criteria

- User can scan or import a document.
- Guided framing helps the user position the phone for document scanning.
- SmartCess produces and reads aloud a plain-language summary.
- SmartCess identifies key points, deadlines, and amounts where present.
- User can ask follow-up questions, save the summary, and create a reminder.

10.5 Appointment Assistant

Appointment Assistant helps users understand, organize, and prepare for appointments.

User flow

- User says: “Add my clinic appointment” and scans a letter, imports a document, or speaks details.
- If scanning, guided framing is used.
- Muse Spark extracts appointment information; SmartCess confirms before saving.
- SmartCess creates a calendar event and travel reminder after approval.
- SmartCess prepares useful questions and includes the appointment in Daily Briefing.

Appointment fields

- Title, provider or clinic name, date and time, location, doctor or contact person, preparation instructions, required documents or items, travel reminder time.

Travel reminder

For the initial release, use a user-selected travel duration, such as 30, 60, or 90 minutes before the appointment. Live traffic and map integration are future enhancements.

Acceptance criteria

- User can add an appointment by voice, scan, import, or manual entry.
- Guided framing used when scanning appointment letters.
- SmartCess extracts details and asks for confirmation before saving.
- Calendar event and travel reminder created after confirmation.
- Appointment appears in Daily Briefing; user can ask questions or cancel/change reminders.

10.6 Routine Guardian

Routine Guardian helps users manage repeating responsibilities, including medication, deliveries, recurring tasks, and daily check-ins.

User flow

- User says: “Add my medication” and provides name, dose, time, and frequency.
- SmartCess confirms and saves the routine.
- At the scheduled time, SmartCess sends a reminder; user confirms completion.
- SmartCess records completion and includes the routine in Daily Briefing.

Routine types and medication rules

- Medication, delivery, recurring task, custom routine.
- Store medication name, dose, time, frequency; send reminders; track completion; alert for missed doses.
- Never diagnose conditions, recommend medication changes, or provide medical advice.

Acceptance criteria

- User can add a routine by voice or touch.
- SmartCess schedules reminders and sends notifications.
- User can mark a routine complete; missed routines tracked.
- Routines appear in Daily Briefing; user can ask questions.

11. Screen inventory and navigation

Navigation map

```
Home / Today
■■■ Talk
■■■ Today / Daily Briefing (max 3 urgent items + See all)
■■■ Scan Bill (primary action)
■■■ More
■   ■■■ Bills
■   ■   ■■■ Bill Result
■   ■   ■■■ Bill Comparison
■   ■   ■■■ Reminder
■   ■■■ Documents
■   ■   ■■■ Scan Document (with guided framing)
■   ■   ■■■ Import PDF
■   ■   ■■■ Document Result
■   ■■■ Appointments
■   ■   ■■■ Add Appointment
■   ■   ■■■ Appointment Detail
■   ■   ■■■ Travel Reminder
■   ■■■ Routines
■   ■   ■■■ Add Routine
■   ■   ■■■ Routine Detail
■   ■   ■■■ Complete Routine
■   ■■■ Settings
```

Screen requirements

Screen	Required elements
Home / Today	SmartCess title, Talk, Today card (max 3 urgent items + See all), Scan Bill (primary), More, Settings
More sheet	Bills, Documents, Appointments, Routines, Settings
Voice	Listening indicator, live transcript, Stop, Cancel, error message
Camera	Camera preview, real-time framing guidance (spoken + haptic), Capture (enabled when framed), Retake, Cancel, processing indicator
Bill Result	Provider, amount, currency, due date, account reference, billing period, required action, summary, confidence warning, Read Aloud, Set Reminder, Save, Rescan, Discard
Bill Comparison	Previous amount, current amount, difference, percentage change, explanation, recommendation
Bills list	Active bills, overdue section, due-soon section, archived section, bill detail navigation
Reminder	Bill title, due date, reminder options, Confirm, Cancel
Email Briefing	Briefing summary, email list, urgency labels, deadlines, attachments, open email, create reminder
Document Result	Title, document type, summary, key points, deadline, amount, required action, Read Aloud, Save, Ask question, Set Reminder
Appointment Detail	Title, provider, location, time, doctor, preparation notes, questions, travel reminder, calendar action
Routine Detail	Title, type, schedule, next due time, Complete, Pause, Delete
Settings	Meta AI status, speech rate, default reminder time, haptic feedback toggle, privacy explanation, delete data, about

Navigation rules

- Home is the default screen; Talk is always accessible from Home.
- Home shows a maximum of: Talk, one Today card (max 3 items), one primary action (Scan Bill), and one More entry point plus Settings.
- All other modules (Bills, Documents, Appointments, Routines) live behind More.
- Scan Bill opens Camera directly with guided framing active.
- Every briefing item links to its module.
- Every bill can open Reminder directly.
- Every document with a deadline can create a reminder.
- Every appointment can create a travel reminder.
- Back navigation must return to the previous logical screen.

12. Muse Spark prompts and schemas

Prompting principles

- Always instruct Muse Spark to return only valid JSON matching the schema.
- Never allow the model to invent provider, amount, due date, or account information.
- Always request a confidence score and missing fields.
- Use plain-language summaries suitable for spoken output.
- Do not request medical, legal, or financial advice.

Intent classification prompt

```
You are SmartCess, an accessibility-first assistant for blind and
low-vision users.

Classify the user's request and identify missing information.

Supported intents:
- daily_briefing
- scan_bill
- read_document
- email_briefing
- ask_bill_question
- create_reminder
- cancel_reminder
- archive_bill
- add_appointment
- add_routine
- general_question

Rules:
- Return only JSON matching the provided schema.
- If the request is unclear, choose the closest intent and provide
  a clarifying question.
- Never invent an intent.
- Never assume a bill provider, amount, or due date.

User request: "{{transcript}}"
Current date and time: "{{current_date_time}}"
```

Intent schema

```
{
  "intent": "daily_briefing | scan_bill | read_document | email_briefing |
             ask_bill_question | create_reminder | cancel_reminder |
             archive_bill | add_appointment | add_routine | general_question",
  "confidence": 0.0-1.0,
  "missing_information": ["string"],
  "clarifying_question": "string or null"
}
```

Bill extraction prompt and schema

```
Extract the key details from this utility bill.

Rules:
- Use the currency shown on the bill.
- Use ISO date format YYYY-MM-DD.
- If a field is unclear, add it to missing_fields.
- Never invent a provider, amount, due date, account number, or billing period.
- Provide a short, plain-language summary for a blind user.
- Do not provide financial, legal, or medical advice.

Return only JSON matching the provided schema.

{
  "provider": "string",
  "amount": 0.0,
  "currency": "string",
  "due_date": "YYYY-MM-DD",
  "account_reference": "string or null",
  "billing_period_start": "YYYY-MM-DD or null",
  "billing_period_end": "YYYY-MM-DD or null",
  "required_action": "pay_before_due_date | review_bill |
```

```
                    contact_provider | no_action_required",
  "confidence": 0.0-1.0,
  "missing_fields": ["string"],
  "summary": "string"
}
```

Bill comparison prompt and schema

```
Compare this bill with the user's previous bill from the same provider.

Previous bill: provider, amount, currency, due date, billing period.
Current bill: provider, amount, currency, due date, billing period.

Rules:
- Explain the difference in simple language.
- Do not speculate about charges not visible in the bills.
- If the reason for the change is unclear, say so.
- Return only JSON matching the provided schema.

{
  "amount_changed": true/false,
  "difference": 0.0,
  "percentage_change": 0.0,
  "possible_reasons": ["string"],
  "recommendation": "string",
  "confidence": 0.0-1.0
}
```

Email briefing prompt and schema

```
Summarize the user's recent unread email.

For each email: short summary, urgency, deadline if present,
required action if present, attachment presence.
Then create a short spoken briefing.

Rules:
- Use simple language.
- Prioritize urgent and time-sensitive messages.
- Do not speculate.
- Do not recommend replying, deleting, or forwarding automatically.
- Return only JSON matching the provided schema.

{
  "emails": [
    {
      "message_id": "string",
      "sender": "string",
      "subject": "string",
      "summary": "string",
      "urgency": "low | normal | high | urgent",
      "deadline": "YYYY-MM-DD or null",
      "required_action": "string or null",
      "has_attachment": true/false
    }
  ],
  "briefing": "string"
}
```

Document summary prompt and schema

```
Summarize this document for a blind or low-vision user.

Rules:
- Use simple, plain language.
- Identify the document type.
- List the most important points.
- Extract deadline, amount, currency, and required action if present.
- Do not provide legal, medical, or financial advice.
- Clearly identify uncertain or missing information.
- Return only JSON matching the provided schema.

{
  "title": "string",
  "document_type": "string",
  "summary": "string",
  "key_points": ["string"],
  "deadline": "YYYY-MM-DD or null",
  "amount": 0.0 or null,
  "currency": "string or null",
```

```
  "required_action": "string or null",
  "confidence": 0.0-1.0
}
```

Appointment extraction prompt and schema

```
Extract appointment details from this document or user input.

Rules:
- Extract title, provider, location, date, time, doctor, preparation notes.
- Use ISO date and time format.
- Do not invent missing details.
- Return only JSON matching the provided schema.

{
  "title": "string",
  "provider_name": "string or null",
  "location": "string or null",
  "start_time": "YYYY-MM-DDTHH:mm",
  "end_time": "YYYY-MM-DDTHH:mm or null",
  "doctor_name": "string or null",
  "preparation_notes": ["string"],
  "confidence": 0.0-1.0
}
```

Daily briefing prompt and schema

```
Create a short, prioritized daily briefing for a blind or low-vision user.

Use only the provided local data.

Prioritize:
1. Overdue bills.
2. Bills due within 7 days.
3. Appointments today or tomorrow.
4. Urgent email deadlines.
5. Documents with deadlines.
6. Medication and routines due today.

Rules:
- Use simple, respectful language.
- Be concise.
- Do not repeat the same information.
- Do not give medical, legal, or financial advice.
- Suggest only safe next actions.
- Return only JSON matching the provided schema.

{
  "greeting": "string",
  "priorities": [
    {
      "type": "bill | email | document | appointment | routine",
      "title": "string",
      "detail": "string",
      "urgency": "low | normal | high | urgent",
      "suggested_action": "string",
      "reference_id": "string or null"
    }
  ],
  "briefing": "string"
}
```

13. State machines

Bill scanning with guided framing

```
IDLE
 -> LISTENING
 -> INTENT_DETECTED
 -> CAMERA_OPEN
 -> FRAMING_GUIDANCE (loop: ML Kit analysis -> TTS + haptic)
        |
        +-- "Move closer" / "Move left" / "Hold steady" etc.
        |
        v
 FRAMING_GOOD ("Ready. Tap capture or say capture")
 -> IMAGE_CAPTURED
 -> PROCESSING
 -> EXTRACTION_SUCCESS
 -> USER_REVIEW
 -> SAVED

Failure paths:
LISTENING -> NO_SPEECH -> IDLE
LISTENING -> TRANSCRIPTION_FAILED -> IDLE
CAMERA_OPEN -> PERMISSION_DENIED -> PERMISSION_SCREEN
FRAMING_GUIDANCE -> USER_CANCELS -> IDLE
IMAGE_CAPTURED -> PROCESSING -> EXTRACTION_FAILED -> RETAKE_OR_CANCEL
EXTRACTION_SUCCESS -> LOW_CONFIDENCE -> USER_REVIEW
USER_REVIEW -> SAVED | DISCARDED
```

Reminder

```
NOT_CREATED -> REMINDER_OPTIONS -> USER_CONFIRMATION_REQUIRED
-> CONFIRMED -> SCHEDULED -> NOTIFICATION_SENT -> COMPLETED

Failure paths:
REMINDER_OPTIONS -> CANCELLED
USER_CONFIRMATION_REQUIRED -> CANCELLED
CONFIRMED -> SCHEDULING_FAILED -> RETRY_OR_CANCEL
SCHEDULED -> CANCELLED
```

Document processing with guided framing

```
IDLE
 -> INPUT_SELECTED (scan or import)
 -> [if scan] CAMERA_OPEN -> FRAMING_GUIDANCE -> FRAMING_GOOD
 -> IMAGE_CAPTURED
 -> PROCESSING
 -> SUMMARY_READY
 -> USER_REVIEW
 -> SAVED

Failure paths:
INPUT_SELECTED -> PROCESSING -> UNSUPPORTED_FILE -> ERROR
INPUT_SELECTED -> PROCESSING -> EXTRACTION_FAILED -> RETRY_OR_CANCEL
SUMMARY_READY -> LOW_CONFIDENCE -> USER_REVIEW
USER_REVIEW -> SAVED | DISCARDED
```

Appointment

```
IDLE -> INPUT_SELECTED -> [if scan] FRAMING_GUIDANCE -> FRAMING_GOOD
-> PROCESSING -> DETAILS_READY -> USER_REVIEW
-> CONFIRMED -> SAVED -> CALENDAR_EVENT_CREATED -> TRAVEL_REMINDER_SET

Failure paths:
INPUT_SELECTED -> PROCESSING -> EXTRACTION_FAILED -> RETRY_OR_MANUAL_ENTRY
DETAILS_READY -> USER_REVIEW -> CANCELLED
CONFIRMED -> CALENDAR_FAILED -> SAVE_LOCALLY_AND_WARN
```

Routine

```
IDLE -> ROUTINE_DETAILS -> USER_REVIEW -> CONFIRMED -> SCHEDULED
-> REMINDER_SENT -> COMPLETED

Failure paths:
ROUTINE_DETAILS -> CANCELLED
USER_REVIEW -> CANCELLED
SCHEDULED -> MISSED
REMINDER_SENT -> COMPLETED | SNOOZED
```

14. Validation rules

SmartCess must validate every AI extraction before saving or acting on it. It must never silently save uncertain information.

Entity	Field	Rule	Failure response
Bill	Provider	Must not be empty	Ask user to confirm or rescan
Bill	Amount	Must be greater than zero	Ask user to confirm or rescan
Bill	Currency	Must be valid or recognizable	Ask user to confirm
Bill	Due date	Must be valid ISO date; warn if past	Ask user to confirm or enter manually
Bill	Required action	Must match allowed values	Use review_bill as safe default
Bill	Confidence	Below 0.70 requires review	Show warning and require review
Bill	Missing fields	Must be listed	Ask user or offer rescan
Document	Title	Must not be empty	Use “Untitled document”
Document	Summary	Must not be empty	Ask user to retry
Document	Deadline	Must be valid if present	Ask user to confirm
Document	Confidence	Below 0.70 requires review	Show warning
Appointment	Title	Must not be empty	Ask user for title
Appointment	Start time	Must be valid future or current time	Ask user to confirm
Appointment	Confidence	Below 0.70 requires review	Show warning
Routine	Title	Must not be empty	Ask user for title
Routine	Type	Must be medication, delivery, recurring task, or custom	Use custom as default
Routine	Next due time	Must be valid	Ask user to confirm

15. Error handling

Every error must be both visible and spoken. The user must always know what happened and what to do next.

Failure	User-facing message	Technical behavior
Microphone permission denied	SmartCess needs microphone access to hear you.	Show accessible permission screen
Camera permission denied	SmartCess needs camera access to scan bills.	Show accessible permission screen
Notification permission denied	SmartCess needs notification access to remind you.	Show accessible permission screen
No speech detected	I did not hear you. Tap Talk and try again.	Stop voice session
Transcription failure	Sorry, I could not hear that clearly.	Offer touch input
Muse Voice Transcribe timeout	That is taking too long. Please try again.	Retry once
Muse Spark timeout	That is taking too long. Please try again.	Retry once
No document detected	I do not see a document. Point the camera at the bill.	Continue framing guidance
Blurry or unreadable bill	I could not read the bill clearly. Please rescan.	Return to camera with framing guidance
Missing due date	I found the amount but not the due date.	Ask user or allow manual entry
Missing amount	I found the due date but not the amount.	Ask user or allow manual entry
Low confidence	I am not sure I read this correctly.	Require review
No internet	SmartCess cannot reach Meta AI right now.	Use ML Kit OCR + TTS fallback
Reminder scheduling failure	I could not set the reminder.	Log error and show retry
Email not connected	To brief you on email, please connect your email account.	Show connection screen
Unsupported PDF	I cannot read this PDF format.	Offer alternative import

16. Offline fallback

If Meta Model API is unavailable, SmartCess must still provide basic help. Offline mode must be honest about its limitations. Guided framing continues to work offline because it uses on-device ML Kit.

Offline bill flow

- Capture bill using CameraX with guided framing.
- Run ML Kit Text Recognition.
- Display raw extracted text.
- Use Android TextToSpeech to read the text.
- Attempt basic extraction of amount, due date, provider, and currency.
- Clearly say: “Offline mode. Please review the details.”
- Allow user to save a low-confidence bill or retry online later.

Offline document flow

- Capture or import document with guided framing.
- Extract text using ML Kit where possible.
- Read text aloud using Android TTS.
- Do not claim a high-quality AI summary.
- Allow user to save raw text or retry online later.

Offline rules

- Do not claim high accuracy in offline mode.
- Do not silently substitute offline extraction for Meta AI extraction.
- Always label the mode clearly.
- Always allow retry when connectivity returns.
- Guided framing remains fully functional offline.

17. Privacy and safety

Privacy requirements

- Use the camera only when the user requests a scan.
- Framing guidance processes frames on-device only; no frames leave the phone during guidance.
- Avoid continuous background camera monitoring.
- Store bills, documents, reminders, routines, and preferences locally.
- Allow users to delete saved data.
- Request only necessary permissions: camera and microphone.
- Clearly indicate when information is sent to Meta AI.
- Avoid collecting location, contacts, SMS, or call logs.

Safety requirements

- Require explicit user confirmation before creating reminders.
- Require explicit user confirmation before creating calendar events.
- Never make payments automatically.
- Never send email automatically.
- Never submit forms automatically.
- Never make phone calls automatically.
- Never provide medical, legal, or financial advice.
- Never act on low-confidence information without user review.

Privacy screen text

“SmartCess processes your bill or document only when you ask it to. Camera framing guidance happens entirely on your phone. Your reminders and saved information stay on your phone. SmartCess does not make payments or share your data without your action.”

18. Accessibility specification

TalkBack requirements

- Every screen must have a clear heading.
- Every interactive element must have a meaningful content description.
- Focus order must follow the user’s task.
- Related information must be grouped.
- Errors and status changes must use live regions.
- Touch targets must be at least 48dp by 48dp.
- Support large text and high contrast.
- Do not convey information by color alone.
- Every voice command must have a touch equivalent.
- Camera framing guidance must be spoken and optionally haptic.

Focus order

Screen	Focus order
Home	Talk, Today card items, See all, Scan Bill, More, Settings
Camera	Framing guidance (live region), Capture, Retake, Cancel
Bill Result	Provider, Amount, Currency, Due date, Account reference, Billing period, Required action, Summary, Read Aloud, Set Reminder, Save, Rescan, Discard
Reminder	Bill title, Due date, Reminder options, Confirm, Cancel
Document Result	Title, Summary, Key points, Deadline, Read Aloud, Save, Ask question, Set Reminder
Appointment Detail	Title, Provider, Location, Time, Doctor, Preparation, Questions, Travel reminder, Calendar action
Routine Detail	Title, Type, Schedule, Next due time, Complete, Pause, Delete

Voice and touch parity

Voice command	Touch equivalent
“What needs my attention today?”	Tap Today
“Scan my bill”	Tap Scan Bill
“Capture”	Tap Capture (when framing is good)
“Read this bill”	Tap Read Aloud
“Remind me three days before”	Tap Set Reminder, choose 3 days before
“Brief me on my email”	Tap More, then Email
“Read this letter”	Tap More, then Documents
“What is my next appointment?”	Tap More, then Appointments
“What routines do I have today?”	Tap More, then Routines
“Cancel reminder”	Open reminder, tap Cancel

Accessibility test cases

- Enable TalkBack and complete the bill workflow without sighted assistance.
- Verify guided framing gives spoken instructions for each framing condition.
- Verify haptic pulses accompany guidance.
- Verify “Ready” confirmation enables Capture.
- Verify every button has a meaningful label.
- Verify focus order follows the task.

- Verify extraction results are announced.
- Verify errors are announced.
- Verify reminders can be created using TalkBack only.
- Verify large font sizes do not break layout.
- Verify high-contrast mode remains usable.
- Verify camera guidance is spoken.
- Verify offline fallback is announced.
- Verify Home has no more than 6 focus stops before More.

19. Testing plan

Unit tests

- Intent classification parsing.
- Bill extraction parsing.
- Bill validation.
- Reminder date calculation.
- Currency formatting.
- Bill comparison calculation.
- Room DAO queries.
- Repository functions.
- Offline extraction rules.
- Framing engine logic (offset, tilt, lighting thresholds).

Integration tests

- Muse Voice Transcribe mock returns transcript.
- Muse Spark mock returns valid JSON.
- Invalid JSON is rejected.
- Low-confidence bill requires review.
- WorkManager reminder is scheduled.
- Offline fallback activates when API fails.
- Bill comparison uses previous bill correctly.
- Framing engine produces correct guidance for mock frames.

UI and accessibility tests

- Home screen navigation.
- Talk button launches voice flow.
- Scan Bill launches camera flow with framing guidance.
- Framing guidance announced via live region.
- Bill Result screen announces fields.
- Reminder confirmation creates a reminder.
- Error states appear accessibly.
- TalkBack-only completion of bill workflow.
- Large font and high-contrast testing.
- Home screen shows max 3 urgent items and More sheet works.

Manual test cases

- Scan a clear printed electricity bill with guided framing.
- Test “move closer” guidance by holding phone far from bill.
- Test left/right guidance by offsetting the bill.
- Test “ready” confirmation when bill is well-framed.
- Scan a blurry bill.

- Scan a bill with missing due date.
- Scan a bill from a provider with a previous bill.
- Create a reminder three days before due date.
- Receive reminder notification.
- Ask “When is my electricity bill due?”
- Import a PDF letter and request summary.
- Add a medication routine and complete it.
- Enable airplane mode and verify offline fallback including framing guidance.

20. Ten-day build plan

Day	Work	Deliverable
Day 1	Project setup: Kotlin, Compose, Hilt, Room, WorkManager, CameraX, ML Kit, Meta Model API client, navigation, empty screens	App opens and navigates accessibly
Day 2	Muse Voice Transcribe integration, microphone permission, voice command capture, transcript state	User can speak a command
Day 3	Muse Spark intent routing, structured output, tool-calling skeleton	SmartCess understands basic commands
Day 4	CameraX bill capture, ML Kit framing detection, framing engine, TTS + haptic guidance	User gets real-time framing guidance
Day 5	Muse Spark bill extraction, structured JSON, validation	SmartCess extracts bill details
Day 6	Bill Result screen, TalkBack grouping, Read Aloud, Save, Rescan, Discard	User can review and save a bill
Day 7	Bill comparison, archive, bill list, bill Q&A;	SmartCess compares and answers bill questions
Day 8	Reminder creation, WorkManager notification, Daily Briefing	Reminders and briefing work
Day 9	Email Briefing, Document Reader, Appointment Assistant, Routine Guardian	All five modules functional
Day 10	Accessibility polish, error handling, offline fallback, demo video, README, release package	Release-ready package

Priority rule

If time runs out, protect this order:

- Daily Briefing must work.
- Bill Watcher must work end to end, including guided framing.
- Document Reader must work for a single-page image or PDF.
- Appointment Assistant must work for manually added or scanned appointments.
- Routine Guardian must work for medication and recurring tasks.
- Email Briefing may use a local demonstration inbox if full email integration is not ready.

Module priorities

Priority	Module	Release level
P0	Daily Briefing	Fully working
P0	Bill Watcher	Fully working end to end, including guided framing
P1	Document Reader	One-page image/PDF workflow with guided framing
P1	Routine Guardian	Medication and recurring task
P2	Appointment Assistant	Manual or scanned appointment and reminder
P2	Email Briefing	Local demonstration inbox first
P3	SAM 3.1 framing refinement	Post-MVP enhancement

21. Definition of done

Daily Briefing

- User can request briefing by voice and touch.
- Briefing includes bills, appointments, email, documents, and routines.
- Briefing is spoken aloud.
- User can open items.
- Briefing handles empty data gracefully.

Bill Watcher

- User can scan a bill with guided framing.
- Muse Spark extracts structured details.
- SmartCess validates extraction.
- User can review with TalkBack.
- User can compare with previous bill.
- User can save, archive, or discard.
- User can create a reminder.
- Notification fires.

Guided Scanning

- User receives spoken guidance for each framing condition.
- Haptic pulses accompany guidance.
- “Ready” confirmation enables Capture.
- Guidance works offline.
- No guidance spam.

Email Briefing

- User can request briefing.
- Emails are summarized.
- Urgency and deadlines identified.
- User can open an email summary.
- User can create a reminder.
- No irreversible email action occurs automatically.

Document Reader

- User can scan or import a document with guided framing.
- SmartCess produces a summary.
- Summary is read aloud.
- Deadlines and amounts extracted where present.
- User can ask follow-up questions.
- User can save summary and create reminder.

Appointment Assistant

- User can add appointment.

- SmartCess extracts details.
- User confirms before saving.
- Calendar event created after approval.
- Travel reminder created after approval.
- Appointment appears in Daily Briefing.

Routine Guardian

- User can add medication, delivery, or recurring task.
- SmartCess schedules reminders.
- User can complete routines.
- Missed routines tracked.
- Routines appear in Daily Briefing.

Home simplicity

- Home shows max 3 urgent items.
- Single primary action (Scan Bill).
- All other modules behind More.
- TalkBack focus stops on Home ≤ 6 before More.

22. Demonstration video — 3 minutes

The hackathon requires a 3-minute demo video. This is one of the main things judges score. Use the following structure to cover problem, solution, Meta AI usage, and working product.

3-minute script

Time	Content	Judging criterion
0:00–0:20	Problem + personal story: “Blind users often need sighted help for bills, email, appointments. I’m building SmartCess with Hillarry, a blind co-designer, to change that.”	Theme fit, impact
0:20–0:40	Daily Briefing: User says “What needs my attention today?” SmartCess gives a spoken, prioritized briefing.	Muse Spark, creativity
0:40–1:20	Guided Bill Scan: “Scan my electricity bill” → real-time framing guidance (“Move closer… move slightly left… Ready”) → capture. This is the standout feature.	Technical ambition, accessibility
1:20–1:50	Extraction + comparison: Muse Spark extracts provider/amount/due date; compares with previous bill (“RM12 higher than last month”).	Muse Spark depth
1:50–2:10	Reminder: “Remind me 3 days before” → confirmation → notification fires.	Working product
2:10–2:35	Quick montage: Document Reader (8s), Appointment Assistant (8s), Routine Guardian (medication reminder, 9s).	Full module coverage
2:35–2:50	TalkBack moment: show the app being used entirely with TalkBack.	Accessibility depth
2:50–3:00	Close: “SmartCess turns inaccessible life admin into independent action. Built with Meta’s Muse Voice Transcribe and Muse Spark.”	Meta tooling, theme

Demo rules

- Total length: exactly 3 minutes (do not exceed).
- Show only features/code built during Oct 26 – Nov 4.
- Make clear what is new (guided framing, Muse Spark extraction, etc.).
- Show Meta AI being used (Muse Voice Transcribe + Muse Spark).
- Use sample/demo data, not real personal bills.
- Record with TalkBack enabled for at least one segment.
- Upload/link the video in the submission form.

Rules compliance checklist

- All code/features built during Oct 26 – Nov 4.
- Public GitHub repository.
- Demo video is 3 minutes and linked in submission.
- Every team member added to submission.
- Project fits “Build for Your World” theme.
- No banned project type (not medical/health advice, not basic RAG, not simple image analyzer).
- Routine Guardian positioned as “medication/routine reminders”, never “health advice”.
- No use of code/data/assets without rights.
- Submitted by Nov 4, 9:00 PM ET.

23. Portal submission checklist

When submitting the project, include the following:

- Project name: SmartCess.
- One-line description: SmartCess is a voice-first accessibility assistant that helps blind and low-vision users understand bills, email, documents, appointments, and routines—and turn them into safe, confirmed actions.
- Selected theme: Build for Your World.
- Selected track: Indie Developer Track.
- 3-minute demo video link (required).
- Public GitHub repository link.
- README link or content.
- Architecture explanation.
- Meta AI tools used: Muse Voice Transcribe, Muse Spark, SAM 3.1 (optional), Muse Image (optional), Muse Code.
- Target users: blind and low-vision Android users.
- Key workflow: Daily Briefing and Bill Watcher with guided scanning.
- Safety statement: SmartCess prepares actions but requires explicit user approval. Does not provide medical, legal, or financial advice.
- Privacy statement: local-first storage, on-device framing guidance, minimal permissions.

24. UI and visual design specification

Design principles

- Calm and trustworthy: reduce visual noise; use clear hierarchy.
- High contrast: meet accessibility contrast requirements in all themes.
- Voice-first but touch-complete: every voice action has a visible touch control.
- TalkBack-first: semantic structure, focus order, and labels drive the layout.
- Forgiving: make it easy to retry, undo, or discard.
- Privacy-visible: clearly indicate when data is processed or sent to Meta AI.
- Progressive disclosure: Home shows only what matters now; everything else behind More.

Colour palette

Token	Light mode	Dark mode	Usage
colorPrimary	#0B3D91	#8AB4F8	Primary buttons, active tab, links
colorOnPrimary	#FFFFFF	#062E6F	Text/icons on primary
colorSurface	#FFFFFF	#121212	Cards, sheets, screens
colorBackground	#F6F8FB	#000000	Screen background
colorOnSurface	#1A1C1E	#E3E2E6	Primary text
colorSuccess	#1B7F3B	#7BD88F	Saved, completed, framing good
colorWarning	#B26A00	#FFB74D	Low confidence, review needed, framing adjustment
colorError	#B3261E	#F2B8B5	Errors, destructive actions
colorFocus	#FFD54F	#FFD54F	Focus indicator

Typography

Style	Font	Size / line height	Usage
Display	Roboto / system	28sp / 34sp	App title on Home
Heading	Roboto / system	22sp / 28sp	Screen titles
Section	Roboto / system	16sp / 22sp	Card and section titles
Body	Roboto / system	14–16sp / 20–24sp	Main content
Caption	Roboto / system	12sp / 16sp	Timestamps, helper text
Minimum	Roboto / system	12sp	Never smaller than 12sp

Components

Component	Specification
Primary button	Filled, 48dp minimum height, 8dp corner radius, clear label, loading and disabled states
Secondary button	Outlined or tonal, 48dp minimum height, used for Cancel, Rescan, Discard
Card	Surface color, 12dp corner radius, 8dp padding, 1dp border in dark mode
List item	Title, subtitle, trailing action, 56dp minimum height, divider optional
Input field	Outlined text field, label always visible, helper and error text below
Dialog	Title, message, primary and secondary actions, focus trapped, dismissible
Banner	Used for offline mode, low confidence, and permission warnings; announced via live region
Chip	Used for urgency and status: Urgent, High, Normal, Low, Paid, Archived
Bottom sheet (More)	Contains Bills, Documents, Appointments, Routines, Settings; opens from Home
Framing overlay	Corner brackets around detected document; color shifts from warning to success when framed; announced via live region

Screen layout rules

- Use a single-column layout on phones.
- Keep primary action at the bottom of the screen where possible.
- Group related fields into cards: Bill details, Amount, Due date, Actions.
- Use 16dp horizontal margins and 8dp vertical spacing between related items.
- Keep the most important information in the first two focus stops.
- Home must show a maximum of: Talk, one Today card (max 3 items), one primary action, More, Settings.

Key screen wireframes

```
HOME (simplified)
+--------------------------------+
| SmartCess                      |
| [Talk]                         |
|                                |
| Today                          |
| • Bill due in 3 days           |
| • Appointment at 2 p.m.        |
| • Medication at 8 p.m.         |
| [See all]                      |
|                                |
| [Scan Bill]                    |
|                                |
| [More]              [Settings] |
+--------------------------------+

CAMERA (guided framing)
+--------------------------------+
| ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ |
| ■  [detected doc bounds]   ■   |
| ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ |
|                                |
| "Move slightly left"           |
|                                |
| [Capture]  [Retake] [Cancel]   |
+--------------------------------+

BILL RESULT
+--------------------------------+
| Electricity Bill               |
| Provider: TNB                  |
| Amount: RM120.00               |
| Due date: 2026-11-05           |
| [Read Aloud]                   |
| [Set Reminder]                 |
| [Save] [Rescan] [Discard]      |
+--------------------------------+
```

Dark mode and contrast

- Support light and dark themes.
- Maintain at least 4.5:1 contrast for normal text and 3:1 for large text.
- Never rely on colour alone; pair colour with text labels or icons.
- Focus indicator must be visible in both themes.

25. Onboarding and first-run experience

Goal

Help a new user understand SmartCess, grant necessary permissions, and complete a first successful action within two minutes.

Onboarding steps

- Welcome screen: explain SmartCess in one sentence and offer “Get started” and “Skip”.
- Accessibility check: confirm TalkBack is enabled and explain how SmartCess supports it.
- Permission requests: microphone, camera, notifications — one at a time with clear rationale.
- Voice setup: quick test of Muse Voice Transcribe with a sample command.
- First bill scan: guided scan of a sample or real bill, demonstrating framing guidance.
- Privacy explanation: what is stored locally and what is sent to Meta AI.
- Finish: land on Home with Daily Briefing visible.

Onboarding rules

- Each step must be skippable except essential permissions.
- Every step must be completable with TalkBack only.
- If a permission is denied, explain the impact and offer a retry path.
- Never block the user from exploring the app.

Acceptance criteria

- New user can complete onboarding in under two minutes.
- All onboarding screens are TalkBack accessible.
- User can skip optional steps.
- User lands on Home with a clear next action.

26. API security, configuration, and cost controls

API key management

- Never hard-code API keys in source code.
- Store keys in local.properties or environment variables for development.
- Use Android Keystore or EncryptedSharedPreferences for any key stored on device.
- Do not commit keys to version control.
- Support key rotation without app reinstall where possible.

Network security

- Use HTTPS for all API calls.
- Validate TLS certificates.
- Do not log raw bill images, document text, or API keys.
- Sanitize error messages before display.

Rate limits and retries

Scenario	Policy
Transient network error	Retry up to 2 times with exponential backoff
Muse Spark timeout	Retry once, then show accessible error
Rate limit exceeded	Show “Too many requests. Please try again shortly.”
Invalid JSON response	Reject, log safely, retry once with corrected schema
Repeated failures	Fall back to offline OCR + TTS where applicable

Cost controls

- Compress bill/document images before upload; target under 1–2 MB per image.
- Limit PDF pages processed per request in MVP.
- Cache Muse Spark responses for identical requests where appropriate.
- Track API usage per feature to identify expensive flows.
- Disable non-essential AI features (e.g., Muse Image, SAM 3.1) by default.

27. AI quality and evaluation plan

Goal

Ensure Muse Spark outputs are accurate, consistent, and safe before release and after prompt changes.

Evaluation dataset

- Create a test set of at least 20 sample bills (clear, blurry, partial, multi-currency).
- Create at least 10 sample documents (letters, notices, PDFs).
- Create at least 10 sample emails (urgent, normal, with/without deadlines).
- Create at least 10 appointment and routine samples.
- Create framing test cases: centered, off-left, off-right, too far, too close, tilted, dark.

Quality thresholds

Metric	Target
Bill field extraction accuracy	At least 90% on clear bills
Due date accuracy	At least 95% on clear bills
Document summary usefulness	Rated useful by tester in at least 80% of cases
Intent classification accuracy	At least 95% on test set
Structured output validity	100% of responses parse against schema
Hallucination rate	Zero invented provider/amount/due date
Framing guidance accuracy	Correct instruction in at least 90% of test frames

Prompt versioning

- Store prompts in version-controlled files.
- Tag each prompt with a version number.
- Run evaluation suite before merging prompt changes.
- Keep a changelog of prompt changes and their measured impact.

Regression testing

- Automated tests parse Muse Spark mock responses against schemas.
- Manual evaluation on the sample dataset before each release.
- Any drop in accuracy blocks release until fixed.

28. Model fallback strategy

The PDF already defines offline fallback. This section defines what happens when only part of the AI pipeline fails.

Failure	Fallback behavior
Muse Voice Transcribe fails	Offer touch input; user can type or tap instead
Muse Spark intent fails	Default to Home with suggestion to retry or use touch navigation
Muse Spark bill extraction fails	Offer rescan or manual entry of bill details
Muse Spark document summary fails	Show raw OCR text and read aloud; offer retry
Muse Spark email briefing fails	Show list of unread emails without AI summary; offer retry
Structured output invalid	Retry once with corrected schema; if still invalid, use safe default and warn user
Partial extraction (some fields missing)	Save only validated fields; mark others as missing; ask user
ML Kit framing detection fails	Fall back to manual capture with spoken hint “Point the camera at the bill and tap capture”
No connectivity	Full offline OCR + TTS mode; framing guidance still works (on-device)

Rules

- Never silently degrade quality.
- Always inform the user which mode is active.
- Always allow retry when the failed component recovers.

29. Content safety and sensitive information

Sensitive content categories

- Financial: bank details, account numbers, card numbers.
- Medical: diagnoses, medications, test results.
- Legal: court notices, contracts.
- Personal: addresses, ID numbers.

Handling rules

- Do not repeat full account or card numbers in spoken output; mask middle digits.
- Do not provide medical, legal, or financial advice.
- If content is ambiguous or potentially sensitive, summarize cautiously and recommend professional help where appropriate.
- If content contains threats, abuse, or crisis language, provide a calm, supportive message and suggest contacting a trusted person or professional service.
- Never store or transmit more data than needed for the task.

Refusal wording

“I can help you understand this document, but I cannot give medical, legal, or financial advice. Here is a summary of the key information.”

30. Localization

Languages

- English is the default language for the initial release.
- Design strings to be externalized from day one for future languages.
- Priority future languages: Bahasa Malaysia, Chinese (Simplified), Tamil.

Localization rules

- All user-facing strings in strings.xml, not hard-coded.
- Use locale-aware date and currency formatting.
- Prompts must be translatable or language-aware.
- Test with pseudo-localization to catch layout breaks.
- Support RTL layouts if Arabic or similar is added later.

Acceptance criteria

- No hard-coded user-facing strings.
- Date and currency render correctly for the device locale.
- Adding a new language requires no code changes, only string resources.

31. Analytics and diagnostics

Goal

Understand how users use SmartCess and where they fail, without collecting unnecessary personal data.

Events to track

Event	Properties
app_open	timestamp, app_version
onboarding_completed	duration, steps_skipped
voice_command	intent, success, duration
bill_scan_started	timestamp
framing_guidance_given	condition, duration_to_ready
bill_scan_completed	success, confidence, duration
bill_saved	provider, has_reminder
reminder_created	lead_time
reminder_fired	timestamp
reminder_completed	on_time yes/no
document_processed	source_type, success, confidence
email_briefing_requested	email_count, urgent_count
appointment_added	has_calendar, has_travel_reminder
routine_completed	type, on_time
error_shown	error_type, screen
api_call	model, latency, success

Privacy rules

- Analytics must be opt-in.
- Never send bill images, document text, email content, or personal identifiers.
- Aggregate and anonymize where possible.
- User can disable analytics in Settings.

Crash reporting

- Integrate crash reporting (e.g., Firebase Crashlytics) with user consent.
- Strip personal data from crash logs.

32. Performance requirements

Metric	Target
Cold start to Home	Under 2 seconds on mid-range device
Voice command to intent	Under 3 seconds
Framing guidance latency	Under 500ms per instruction
Bill capture to extraction result	Under 30 seconds
Document import to summary	Under 45 seconds for single-page
Daily Briefing generation	Under 10 seconds
Memory usage during scan	Under 300 MB peak
Battery impact	No significant drain from background work
APK size	Under 50 MB download size

Optimization rules

- Compress images before upload.
- Process every 2–3 camera frames for framing, not every frame.
- Use pagination for lists.
- Cache Daily Briefing for the current day.
- Debounce voice input.
- Profile and optimize on a low-end device.

33. Device support matrix

Item	Requirement
Minimum Android version	Android 8.0 (API 26)
Target Android version	Latest stable at release
Screen sizes	Phone portrait primary; tablet secondary
Screen readers	TalkBack (latest stable)
Camera	Any rear camera, autofocus preferred
Microphone	Required for voice features
Haptics	Standard vibrator; stereo haptics optional
Connectivity	Wi-Fi or mobile data; offline mode required
Low-end devices	Must run on devices with 3 GB RAM

Testing devices

- At least one mid-range device for daily testing.
- At least one low-end device for performance validation.
- Emulator for CI and layout testing.

34. Data migration, export, and deletion

Database versioning

- Use Room schema versioning.
- Write migration for every schema change.
- Test migrations from all supported prior versions.
- Never lose user data during update.

Data export

- Allow user to export bills, documents, reminders, routines as JSON or CSV.
- Export must be initiated explicitly by user.
- Export file stored in user-selected location.

Data deletion

- User can delete individual bills, documents, reminders, routines.
- User can delete all data in Settings.
- Deletion is permanent after confirmation.
- Deleted data is not recoverable.

35. Accessibility conformance

Standard

SmartCess targets WCAG 2.1 Level AA for digital accessibility, adapted for mobile.

Key requirements

- Contrast ratio at least 4.5:1 for normal text, 3:1 for large text.
- All interactive elements reachable and operable via TalkBack.
- No time limits on user input except voice sessions.
- Content reflows at 200% zoom without loss of function.
- Status messages announced via live regions.
- No content depends solely on colour, shape, or position.
- Camera framing guidance available as speech and haptics.

Testing

- Manual TalkBack walkthrough of every core flow.
- Automated accessibility scanner (e.g., Accessibility Scanner) run on all screens.
- Contrast checked with a contrast checker in both themes.
- Accessibility issues block release.

36. Privacy policy and compliance

Privacy policy contents

- What data SmartCess collects: usage analytics (opt-in), crash reports (opt-in).
- What data stays local: bills, documents, reminders, routines, preferences, framing frames.
- What is sent to Meta AI: images/text user chooses to process, voice audio during active session.
- How long data is kept: until user deletes it.
- User rights: access, export, delete.
- Contact for privacy questions.

Compliance

- Comply with applicable data protection laws (e.g., PDPA in Malaysia).
- Obtain consent before analytics or crash reporting.
- Provide clear privacy notice at onboarding and in Settings.
- Do not sell or share personal data with third parties.

37. CI/CD and release process

Build pipeline

- Version control: Git with feature branches.
- CI runs on every push: lint, unit tests, build.
- Nightly build with integration tests.
- Release build triggered manually or on tag.

Release checklist

- All tests pass.
- Accessibility scan clean.
- Manual TalkBack walkthrough complete.
- Performance targets met.
- Privacy policy updated.
- Version number bumped.
- Release notes written.
- Signed APK/AAB generated.
- Play Store listing updated.

Versioning

- Semantic versioning: MAJOR.MINOR.PATCH.
- Version code increments with every release.

38. User research and feedback

Research plan

- Weekly session with Hillarry to test new features.
- Task-based usability testing: “Scan this bill and set a reminder.”
- Collect feedback on clarity, speed, trust, and accessibility.

Feedback channels

- In-app feedback form in Settings.
- Support email.
- Optional post-task micro-survey after key flows.

Iteration rule

- Every major feature must be tested with at least one blind or low-vision user before release.
- Feedback must be triaged within one week.

39. Risk register

Risk	Impact	Likelihood	Mitigation
Muse Spark extraction inaccurate	User acts on wrong info	Medium	Validation, confidence thresholds, user review
API downtime	Core features unavailable	Medium	Offline fallback, retry, status messaging
API cost overrun	Budget impact	Medium	Image compression, caching, usage caps
TalkBack usability issues	Users cannot complete tasks	High	TalkBack-first design, weekly testing
Framing guidance inaccurate	User cannot position phone	Medium	ML Kit tuning, fallback to manual capture, haptic cues
Home screen clutter	Users overwhelmed, missed urgent items	Medium	Progressive disclosure, max 3 urgent items, More sheet
Demo video over 3 minutes	Disqualification risk	Low	Strict 3-minute script, rehearse timing
Privacy breach	Loss of user trust, legal risk	Low	Local-first, encryption, minimal data collection
Low adoption	Product not used	Medium	Onboarding, simple UX, clear value
Scope creep	Delayed release	High	Strict priority rule, cut lines

40. Post-release roadmap

Next 3 months

- Gmail and Outlook integration for Email Briefing.
- Multi-page PDF support.
- Cloud backup and sync (opt-in).
- Improved bill comparison with historical trends.
- SAM 3.1 framing refinement for better document boundary detection.

Next 6 months

- Live traffic and map integration for travel reminders.
- Caregiver sharing with explicit consent.
- Additional languages.
- Wearable or widget support.

Next 12 months

- Bill payment integration with user approval.
- Advanced document understanding (forms, tables).
- Community feedback-driven features.

41. Business model and sustainability

Model

- Free core: Daily Briefing, Bill Watcher, basic Document Reader.
- Premium: unlimited document processing, advanced comparisons, cloud sync, priority support.
- Partnerships: utilities, banks, healthcare providers for integrated bill/appointment data.

Sustainability

- Keep API costs low through caching and compression.
- Monitor cost per active user.
- Seek grants or partnerships focused on accessibility.

42. Team and ownership

Role	Responsibility
Product owner (Maurice)	Vision, priorities, stakeholder communication
Core tester / co-designer (Hillarry)	Usability testing, accessibility feedback
Android developer	App implementation
AI engineer	Meta AI integration, prompts, evaluation
Designer	UI/UX, accessibility design
QA	Testing, accessibility conformance

Decision-making

- Product owner sets priorities.
- Core tester has veto on accessibility blockers.
- Weekly sync to review progress and risks.

43. Updated release checklist

- All sections of this specification implemented or explicitly deferred with rationale.
- UI matches design system.
- Home screen simplified: max 3 urgent items, single primary action, More sheet.
- Guided scanning with spoken + haptic framing feedback implemented.
- Onboarding complete.
- API keys secured.
- AI evaluation thresholds met.
- Offline fallback tested.
- Localization ready.
- Analytics opt-in implemented.
- Performance targets met.
- Device matrix tested.
- Data migration tested.
- WCAG 2.1 AA conformance verified.
- Privacy policy published.
- CI/CD pipeline green.
- Post-release roadmap documented.
- 3-minute demo video recorded and linked.
- Public GitHub repository created.
- All team members added to submission.
- Rules compliance verified (new work only, theme fit, no banned category).

Final instruction to AI coding agent

```
Build SmartCess as a production-quality Android MVP using Kotlin,
Jetpack Compose, Hilt, Room, WorkManager, CameraX, ML Kit, Android TTS,
and Meta Model API.

The primary AI brain is Muse Spark. Muse Voice Transcribe handles
speech-to-text. Muse Spark handles intent classification, multimodal
bill and document understanding, structured extraction, bill comparison,
email briefing, appointment extraction, routine handling, daily briefing,
and tool calling.

CRITICAL FEATURE — Guided Scanning: When the user opens the Camera
for bill or document scanning, run a real-time on-device framing loop
using ML Kit Text Recognition. Analyze detected text bounding boxes and
give ONE simple instruction at a time via TextToSpeech and haptic
feedback:
- No text: "I do not see a document. Point the camera at the bill."
- Too far: "Move closer to the bill."
- Off-left: "Move the phone slightly to the left."
- Off-right: "Move the phone slightly to the right."
- Tilted: "Hold the phone straight over the bill."
- Too dark: "It is too dark. Move to better light if possible."
- Well-framed: "Ready. Tap capture or say capture."
Process every 2-3 frames. Do not spam instructions. Framing must work
offline. Only the final captured image is sent to Muse Spark.

Implement all five SmartCess modules as working features:
Daily Briefing, Bill Watcher, Email Briefing, Document Reader,
Appointment Assistant, and Routine Guardian.

Follow the UI and visual design specification exactly: use the defined
colour tokens, typography scale, component styles, and screen layouts.
Support light and dark themes with WCAG 2.1 AA contrast.

IMPORTANT: The Home screen must be simplified using progressive
```

disclosure. It shows only: Talk, one Today card (max 3 urgent items plus See all), one primary action (Scan Bill), a More bottom sheet (Bills, Documents, Appointments, Routines, Settings), and Settings. Do not place all module buttons directly on Home.

Implement onboarding, API key security, rate limits, cost controls, AI evaluation hooks, model fallback strategy, content safety rules, localization-ready strings, opt-in analytics, performance budgets, device support, data migration/export/deletion, accessibility conformance, privacy policy hooks, CI/CD readiness, user feedback, risk mitigations, and the post-release roadmap structure.

Prioritize TalkBack accessibility, user confirmation for actions, validation of AI output, graceful offline fallback, and reliable end-to-end workflows.

Do not implement payments, automatic email sending, form submission, navigation, or medical advice.

The final deliverable is a working Android app, a 3-minute demonstration video, a complete README, an architecture diagram, a public GitHub repository, and a release package.